

02-741 Week 3

Generative AI for Biomedicine

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Diffusion Models

Diffusion models are state-of-the-art models for (continuous) data generation

- Time series data
- Audio / speech
- 3D objects (coordinates), e.g., Molecule structures

Data Representation

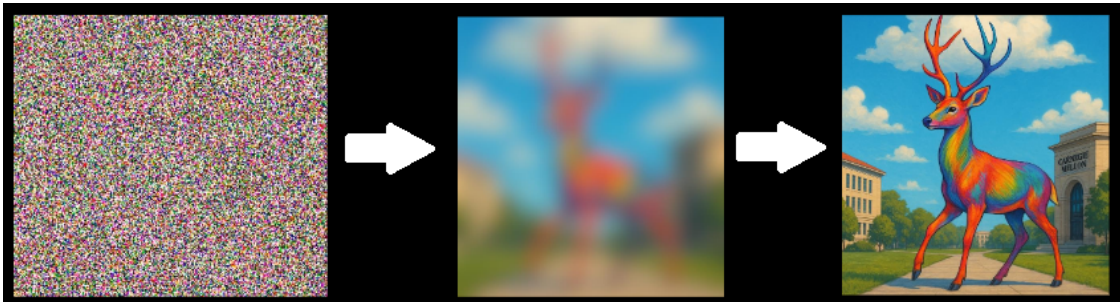
- Images: $x \in \mathbb{R}^{w \times h}$
- Speech: $x \in \mathbb{R}^t$
- Molecular structure: $x \in \mathbb{R}^{N \times 3}$

In each case, we want a probabilistic model $p(x)$ to fit the data and generate samples

Probabilistic Generative Process

- Start from initial distribution $p_T(x_T)$ (e.g., Gaussian $N(0, I)$). This is just noise.
- Generate slightly improved data: $x_{t-1} \sim p_{t-1}(x_{t-1}|x_t)$
- Final data: $x_0 \sim p_0(x_0|x_1)$

The generation process is a Markov chain.



Learning the generative model is equivalent to learning the parameters for each $p_{t-1}(x_{t-1}|x_t)$.

- It is difficult to directly construct a series of probability distributions

Diffusion Model

1. Forward Noising Process

- Synthesize data for training by repeatedly adding noise each step

2. Reverse Generation Process

- Start from complete noise, generate clean data
- Also known as denoising

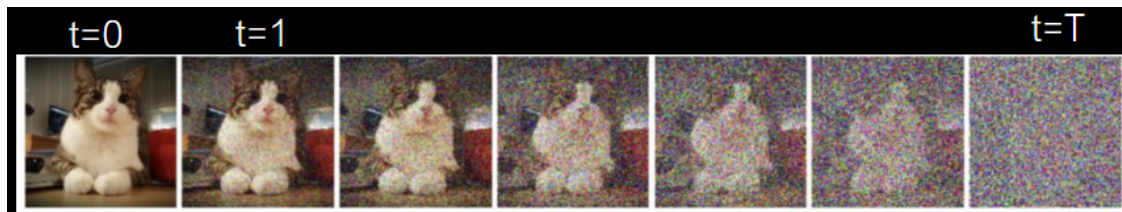
(Forward) Noising Process

- Adding Gaussian noise at each time step t

$$x_t \sim q(x_t|x_{t-1}) = N(\sqrt{\alpha_t}x_{t-1}, \beta_t I), \alpha_t = 1 - \beta_t$$

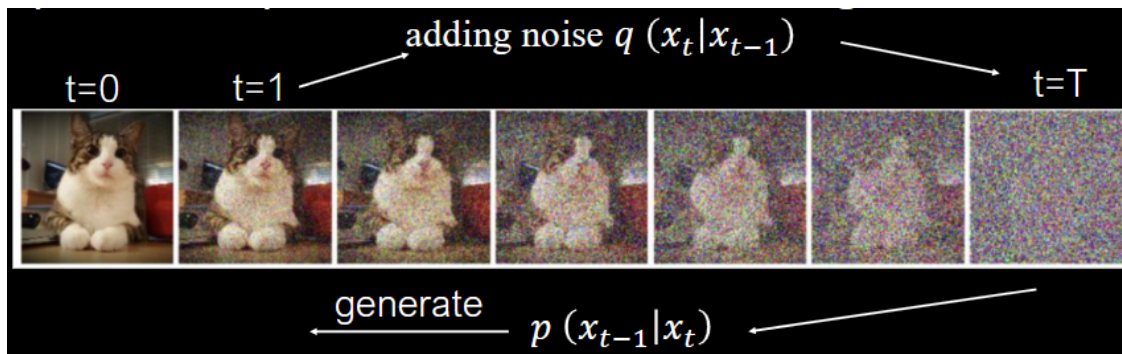
equivalently,

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{\beta_t}\epsilon, \epsilon \sim N(0, I)$$



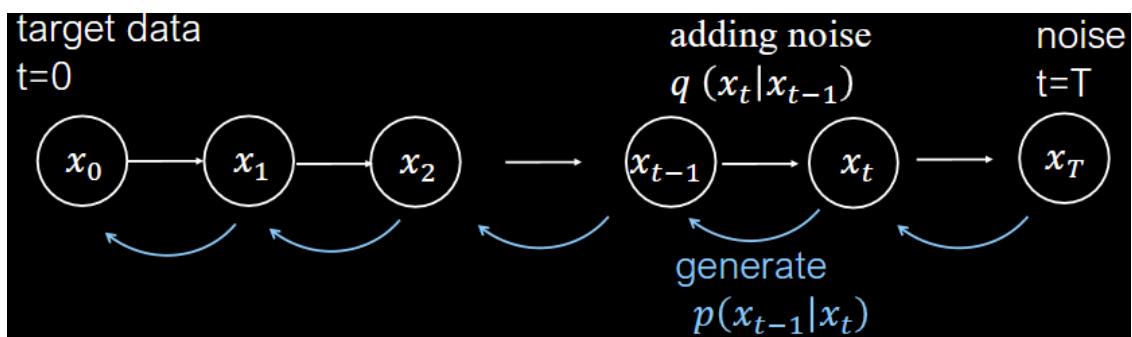
Consequence: with sufficient large T , the result will be Gaussian noise.

(Reverse) Generation / Denoising Process



Learning problem: how to find parameters of p to match the sequence of noised images?

Graphical Model for Diffusion



Denoising Diffusion Probabilistic Models (DDPM)

- Reverse Process (Generation), aka. denoising

$$p_\theta = \prod_{t=1}^T p_\theta(x_{t-1}|x_t)$$
$$p_\theta(x_{t-1}|x_t) = N(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

the μ_θ and Σ_θ functions are predicted by neural network

- Forward Process (Diffusion), aka noising

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1})$$
$$q(x_t|x_{t-1}) = N(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

noising schedule: $\beta_t \in [0, 1]$, and follows a predefined schedule

DDPM Training

For each data sample x_0 (e.g., image), pick random step t

1. sample a noise from standard Gaussian $N(0, I) \rightarrow \epsilon$
2. Noise data at step t : $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$, $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$
3. Estimate noise using neural network: $\hat{\epsilon}_\theta = \text{NN}_{\text{forward}}(x_t, t)$
4. Compute MSE loss: $L = \|\epsilon - \hat{\epsilon}_\theta\|^2$
5. Compute gradient via backward through NN
6. Update model parameters using optimization algorithm (e.g., Adam)

DDPM training is essentially predicting the noise based on image.

DDPM Inference

1. Randomly sample from Gaussian(0, I)
2. For step $t = T$ to 1:
 - Estimate noise using neural network: $\hat{\epsilon}_\theta = \text{NN}_{\text{forward}}(x_t, t)$
 - Generate noise from $N(0, I) \rightarrow \eta$
 - Estimate data mean: $\tilde{\mu}_{t-1} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_\theta \right)$
 - Update data: $x_{t-1} = \tilde{\mu}_{t-1} + \sqrt{\beta_t} \eta$

DDPM inference is essentially iteratively removing predicted noise.

Intuition and Theory

- Training objective:

$$\begin{aligned}
L &= -\log p_\theta(x_0) \\
&\geq -E \left[\log \frac{p_\theta(x_0:T)}{q(x_1:T|x_0)} \right] = L_{elbo} \\
&= \sum_{t=1}^T \text{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t)) + \text{KL}(q(x_T|x_0) || p_\theta(x_T)) - E \log p_\theta(x_0|x_1)
\end{aligned}$$

- L_{elbo} is ELBO loss.
- We can use Jensen's inequality, posterior for linear Gaussian distribution

$$\begin{aligned}
q(x_t|x_0) &= N(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I) \\
\bar{\alpha}_t &= \prod_{s=1}^t (1 - \beta_s)
\end{aligned}$$

$$\begin{aligned}
q(x_{t-1}|x_t, x_0) &= N(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t I) \\
\tilde{\mu}_t &= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) \\
\tilde{\beta}_t &= \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}
\end{aligned}$$

What about $p_\theta(x_{t-1}|x_t)$? Just set:

$$p_\theta(x_{t-1}|x_t) = N(\mu_\theta(x_t, t), \sigma_t^2 I)$$

$$\begin{aligned}
&\text{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t)) \\
&= \text{KL}(N(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t I) || N(\mu_\theta(x_t, t), \sigma_t^2 I))
\end{aligned}$$

let $\sigma_t^2 = \tilde{\beta}_t$, and let

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_\theta(x_t, t) \right)$$

$$\min \|\tilde{\mu}_t - \mu_\theta(x_t, t)\|^2$$

is equivalent to

$$\min \frac{\beta_t^2}{\alpha_t(1 - \bar{\alpha}_t)} \|\epsilon - \hat{\epsilon}_\theta(x_t, t)\|^2$$

Essentially, DDPM is using NN to predict noise given noised data

Summary

Diffusion models:

- is a probabilistic generative model by iteratively removing noise
- construct a sequence of corrupted data by applying Gaussian noises
- use a neural network to estimate the noise given the corrupted data (without knowing the original clean data)
- once estimate the noise, use posterior estimation to remove noise
- can use any suitable NN (e.g., UNet)
- much better than VAE, but much slower

Multi-Objective Drug Molecule Generation via MARS

Drug discovery and development is a long and risky process. The successful launch of a new drug cost \$1.3 Billion on average and more than 10 years.

In silico Drug Discovery

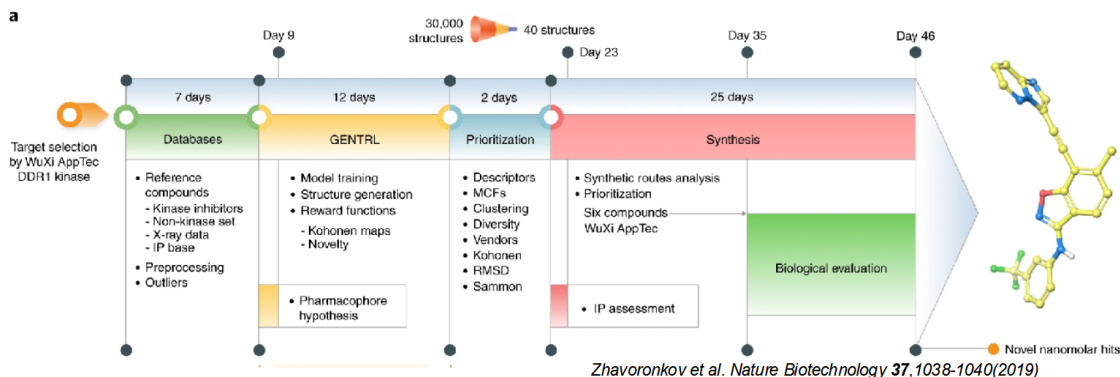
- Using computers to predict drug activities.
- Faster and cheaper.
- But drug candidates are either existing molecules or manually designed ones.

Discovering Treatment for COVID-19

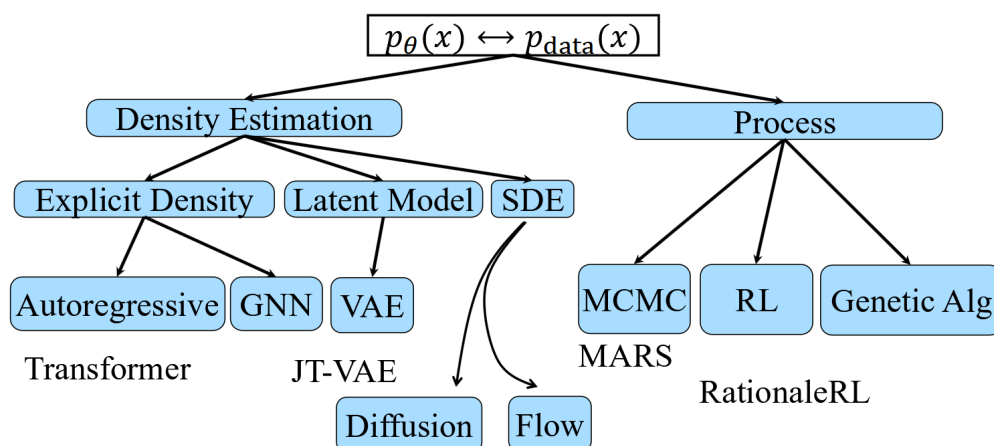
- Paxlovid
- Nirmatrelvir (by modifying prior protease inhibitor)
- Ritonavir: originally for HIV
- Remdesivir: originally for hepatitis C

AI Powered Drug Discovery

- AI can generate drug-like molecules with desired properties.
- Using deep generative model (GENTRL) to design novel and biological active drug candidates in 46 days.

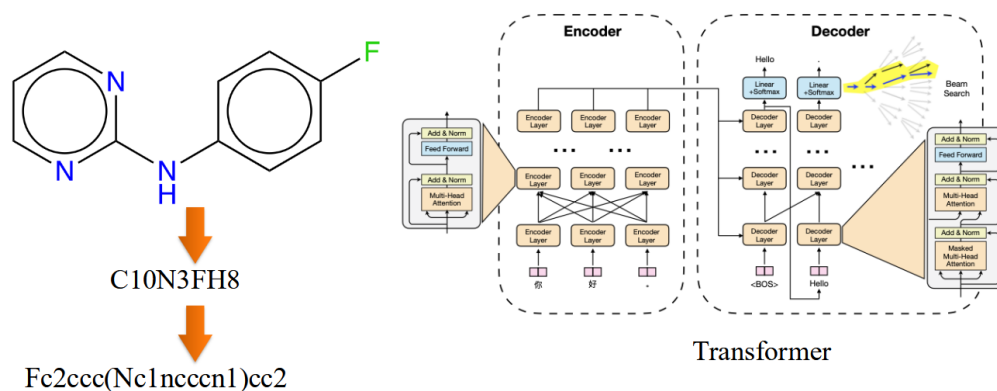


Generative Models for Molecules



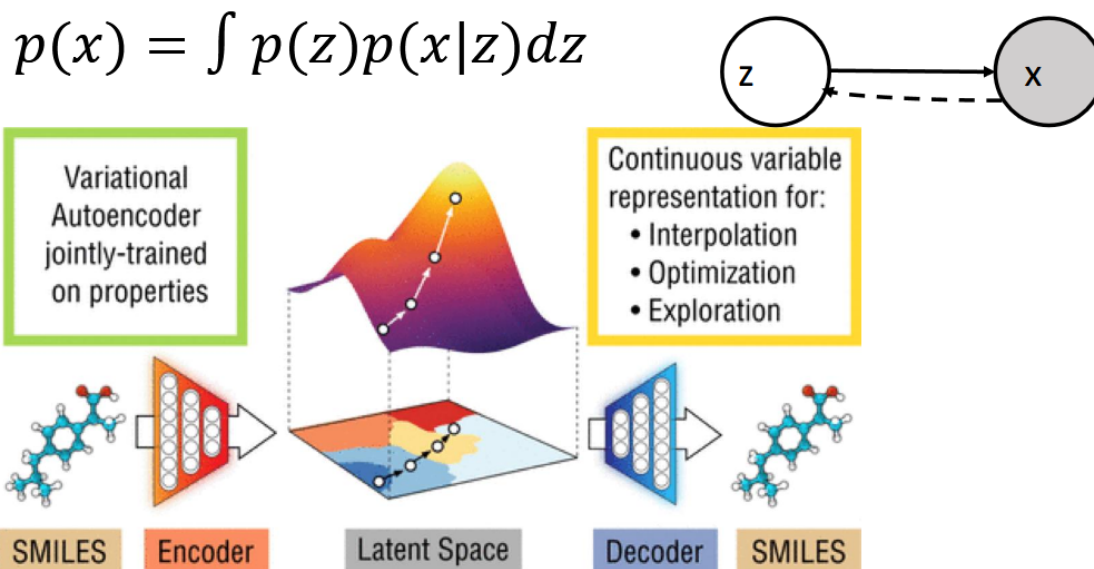
Perspective 1: Molecule Sequence Generation

Transformer based:

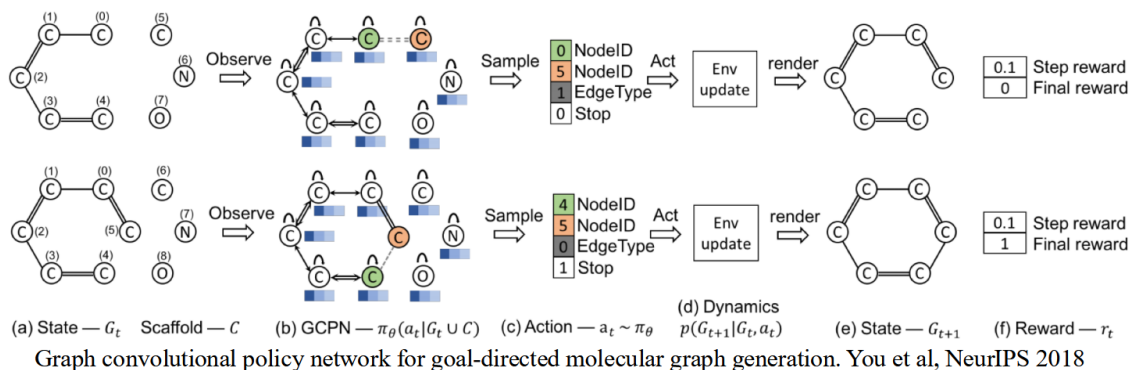


$$p(x) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \dots$$

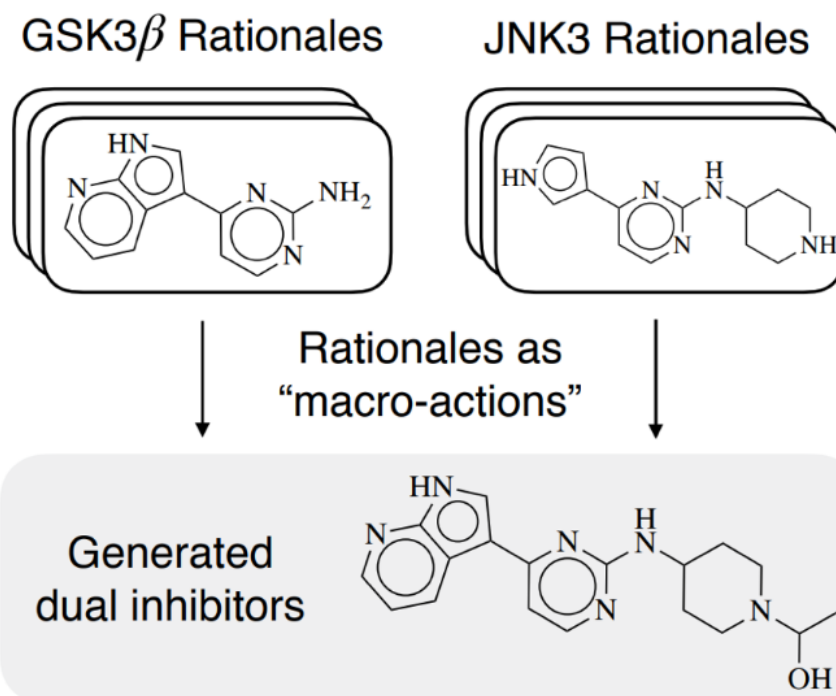
Generation from Latent Space:



Reinforcement-Learning based:

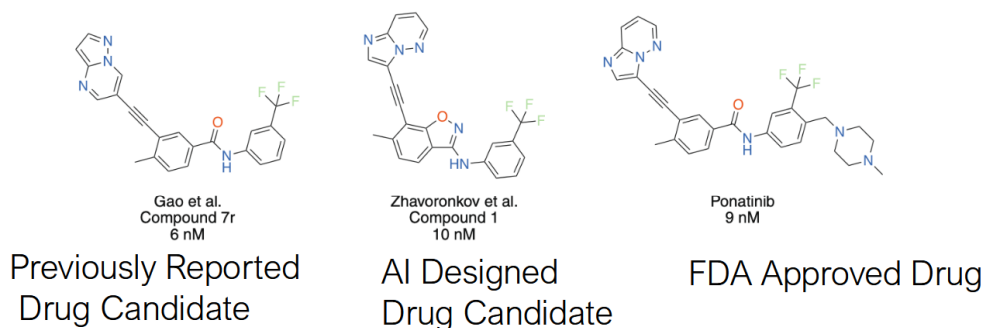


RationaleRL



Challenges

- Current generative models can only find molecules with high similarities comparing to existing drugs / potential drug candidates.



Walters et al. Nature Biotechnology 38, 143–145(2020)

Practical Drug Discovery

Broader exploration of chemical space

- Chemical space is huge but under-explored.
- About 85% drug candidates fail at clinical trials
- Finding diverse drug candidates can improve the success rate in clinical trials.

Multi-Objective Optimization for Drug Discovery

- Successful drugs need to meet multiple requirements, e.g., bioactivity, safety, and etc.
- In drug discovery, these properties should also be optimized.

Goal: Finding Diverse Molecules Satisfying Multiple Objectives

- Satisfy multiple properties with high scores
- Produce diverse and novel molecules
- Does not rely on lab measured data

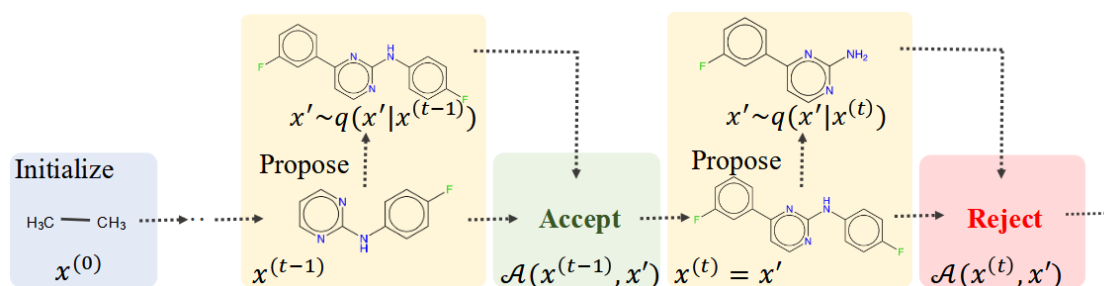
MARS: Markov Molecule Sampling

- MCMC sampling from

$$\pi(x) = s_1(x) \circ s_2(x) \circ s_3(x) \circ \dots \circ s_K(x)$$

- Where each $s_1, s_2, \dots$ represent desired properties.
- $s_k(x)$ is a scoring function for a given desired property.
 - * QED: drug-likeness
 - * SA: possibility to synthesize
- $\pi(x)$ is the unnormalized distribution over molecule space.

MARS does **Iterative Graph Editing**.



- Start from initial molecule (e.g., Ethane)
- For each step, propose a new molecule x' by modifying existing one, $x' \sim q(x'|x^{(t-1)})$
- Accept wrt ratio:

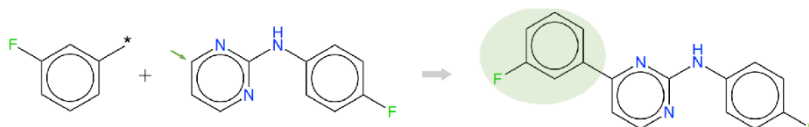
$$\mathcal{A}(x, x') = \min \left\{ \frac{\pi^\alpha(x')q(x|x')}{\pi^\alpha(x)q(x'|x)} \right\}$$

Adaptive Proposal Learning in MARS

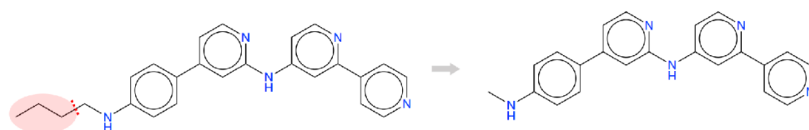
- MCMC sampling
- Markov-chain annealing scheme to find optimal samples
- Adaptive MCMC proposal: MPNN learned on sampled molecular graphs

Molecular Graph Editing

- Adding fragment



- Deleting fragment



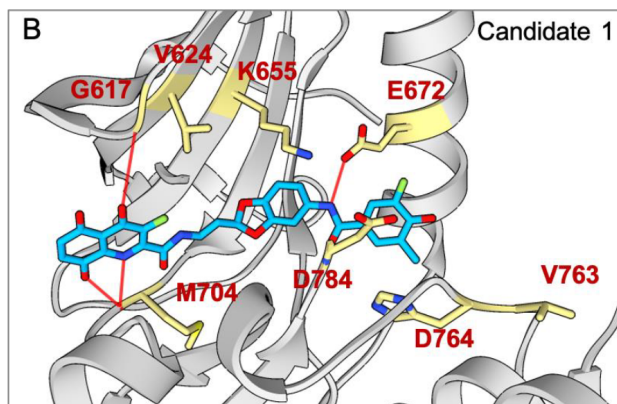
Learning Proposal Network (MPNN)

Modeling editing actions as node, edge, and graph prediction problems

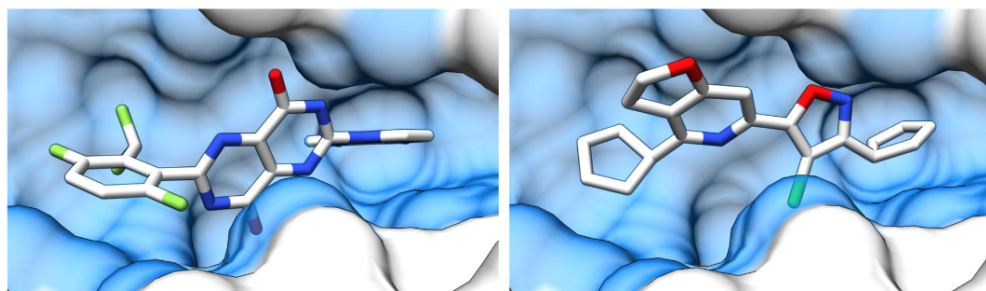
$$\begin{aligned} \mathbf{h}_b^{\text{edge}} &= \text{Concat}(\mathbf{h}_v^{\text{node}}, \mathbf{h}_w^{\text{node}}) \in \mathbb{R}^{2d} \\ \mathbf{h}^{\text{graph}} &= \text{MaxPooling}(\{\mathbf{h}_u^{\text{node}}\}) \in \mathbb{R}^d \\ p_{\text{add}} &= \text{Softmax}(\{\text{MLP}_{\text{node}}(\mathbf{h}_u^{\text{node}})\}) \longrightarrow \text{Prob. adding} \\ p_{\text{frag}} &= \text{Softmax}(\text{MLP}_{\text{graph}}(\mathbf{h}^{\text{graph}})) \longrightarrow \text{Prob. adding} \\ p_{\text{del}} &= \text{Softmax}(\{\text{MLP}_{\text{edge}}(\mathbf{h}_b^{\text{edge}})\}) \longrightarrow \text{Prob. deleting} \end{aligned}$$

MARS with Docking Score

Grade MARS molecules by how well it binds to protein targets



Integrating 3D Structures



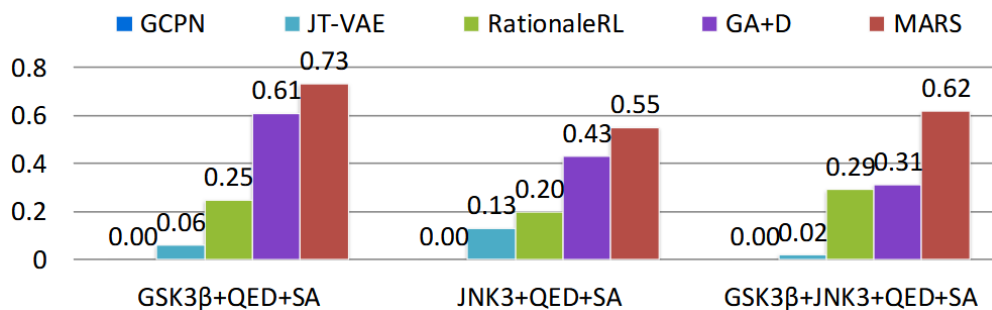
Vina: -12.49 QED: 0.58 SAScore: 0.63

Vina: -12.41 QED: 0.80 SAScore: 0.68

Adaptive Proposal Training

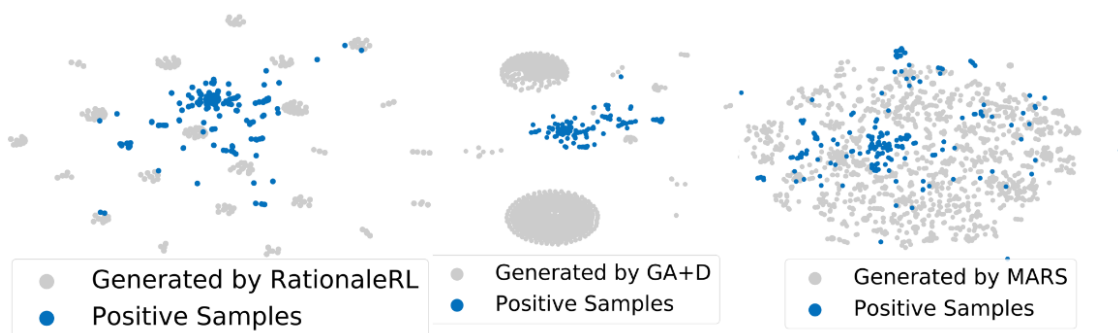
```
Initialize molecules  $x_i^{\{0\}}$  for  $i = 1$  to  $N$ 
Initialize molecular editing model  $M_{\theta}$ 
Create an empty editing model training dataset  $D = \{\}$ ;
for  $t = 1, 2, \dots$ , do
  for  $i = 1, 2, \dots, N$ , do
    calculate probability distributions  $(p_{\text{add}}, p_{\text{frag}}, p_{\text{del}}) = M_{\theta}(x_i^{\{t-1\}})$ 
    Sample a candidate molecule  $x'$  from the proposal distribution
       $q(x' | x_i^{\{t-1\}})$  defined with probability distributions  $p_{\text{add}}, p_{\text{frag}}, p_{\text{del}}$ 
    if  $u < A(x_i^{\{t-1\}}, x')$  where  $u \sim U_{\{[0, 1]\}}$  then
      Accept the candidate molecule  $x_i^{\{t\}} = x'$ 
    else
      Refuse the candidate molecule  $x_i^{\{t\}} = x_i^{\{t-1\}}$ 
    end
    if the candidate improves, i.e.,  $\pi(x') > \pi(x_i^{\{t-1\}})$  then
      adding the editing record  $(x_i^{\{t-1\}}, x')$  into the dataset  $D$ 
    end
  end
  Update model  $M_{\theta}$  with the current dataset  $D$  in a MLE manner
end
```

MARS Generates Better Molecules

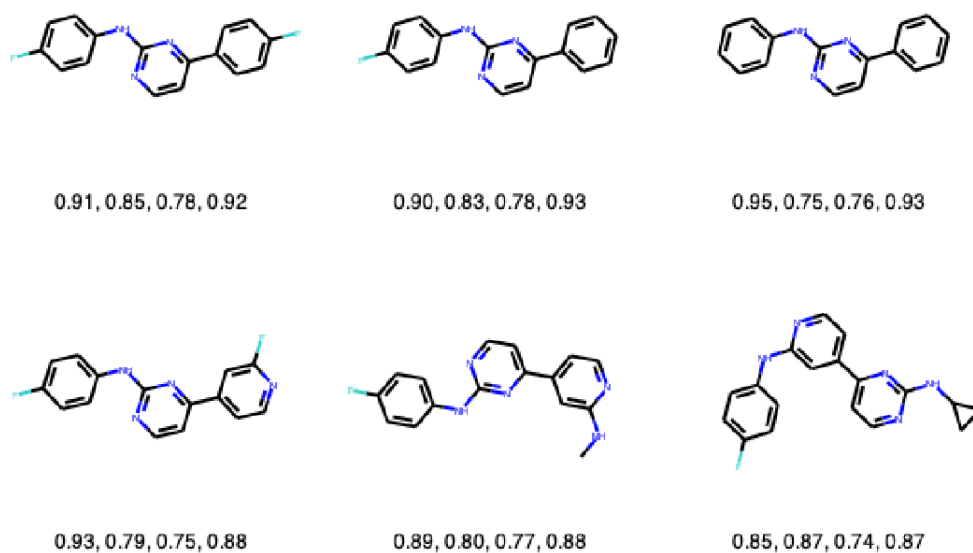


- Generation Objectives
 - GSK3 β , JNK3: biological inhibition
 - QED: drug-likeness score
 - SA: synthetic accessibility score
- Evaluate with Product score = success_rate $\times$ novelty $\times$ diversity

MARS Explores Large Chemical Space



Novel Compounds Found



Key Takeaways

- Challenges in Drug discovery: multi-objective, novel and diverse, lack of data
 - Modeled as a molecule generation problem
 - * Based on MCMC sampling
 - * Self-adaptive proposal trained on the fly $\Rightarrow$ no need for data
 - * Generates better molecules and explores larger chemical space
 - $\Rightarrow$ can discover novel and diverse drug-like molecules
 - Challenges remaining:
 - * More properties
 - * Larger molecule, peptide, protein

Decoding Single-Cell Spatial Genomics and Epigenomics with AI/ML

Introduction

Genome function in different cell types

- Numerous different cell types with distinct functions in our body
- Same genome but different gene regulation and epigenome
- Development of single cell technology provides closer look at cell types and cell states

Multiscale cellular structure and function

- Single-cell 3D epigenome and gene regulation
- Cellular spatial organization and interaction
- Graph / hypergraph neural network, self-supervised, latent embedding, metagene, attention, (foundation model)

Large-scale genome organization in the nucleus [FILL 4]

Hi-C Contact Map

[FILL 5]

Multiscale 3D genome organization

[FILL 6]

Challenges for Single-cell 3D genome analysis using scHi-C data

- High dimensionality
- Sparse and noisy data
- Multiscale 3D genome features

What do 3D genome look like in **single cells**, and how do multiscale 3D features vary in individual cells and different cell types?

Hypergraphs for Higher-Order Interactions

- Hypergraphs are used to represent higher-order interactions
 - Example: events (human, location, activity)
- A **hypergraph** $H = (V, E)$
 - V/E : the set of nodes / **hyperedges**
 - $e \in E$: a hyperedge connects two or more nodes
 - $\forall e \in E, |e| = k$, H is a k -uniform hypergraph
- **Hyper-SAGNN** - hypergraph representation learning
 - Input: a hypergraph with features for each node (non-uniform heterogeneous hypergraph)
 - We aim to:
 - * Learn embeddings for the nodes in the hypergraph
 - * Learn to predict the existence of hyperedges given the node embeddings

[FILL 8, rearrange]

Higashi - modeling scHi-C data as a hypergraph

[FILL 9] [FILL 10]

Higashi separates complex cell types in the human prefrontal cortex

- Higashi embeddings separate neuron subtypes using only scHi-C part of sn-m3c-seq (data from Lee et al. *Nat Methods* 2019)
- Cell type-specific 3D chromatin structures near marker genes

[FILL 11, rearrange]

Mechanisms of Genome Folding

[FILL 12, all]

Patterns of Chromosome Spatial Segregation based on Hi-C data

- Chromosomes are segregated into A and B compartments
- High-coverage Hi-C in GM12878 identified **subcompartments** by clustering inter-chromosomal contact maps

[FILL 13]

SNIPER - Inferring Hi-C Subcompartments

- Autoencoder compresses sparse inter-chromosomal Hi-C contacts into embeddings and imputes missing contacts
- A classifier then uses embedded interchromosomal Hi-C data to predict subcompartments

scGHOST: Identifying Single-Cell 3D Genome Subcompartments

- scHi-C contact maps as **graphs**
- Cell embeddings define k-NN cells. **Genomic locus embeddings** lead to subcompartments
- A unique **random sampling** procedure that filters noise in imputed scHi-C data

[FILL 15] [FILL 16]

- Sampling based on both first-order random walks and second-order random walks
- Graph node embedding using NN

Single-Cell Subcompartments of Human Prefrontal Cortex

[FILL 17]

Spatial Organization of Genomes and Cells

[FILL 18]

Spatial Transcriptomic Technologies

Imaging-based - single cell resolution

- STARmap / STARmap PLUS (Wang et al. *Science* 2018 / Zeng et al. *Nat Neurosci* 2023)
- seqFISH+ (Eng et al. *Nature* 2019)
- MERFISH / Vizgen MERSCOPE (Moffitt et al. *Science* 2018)

Sequencing-based - full transcriptome

- Visium (Stahl et al. *Science* 2016)
- Slide-seq / Slide-seq V2 (Rodriques et al. *Science* 2019 / Stickels et al. *Nat Biotech* 2021)
- Stereo-seq (Chen et al. *Cell* 2022)